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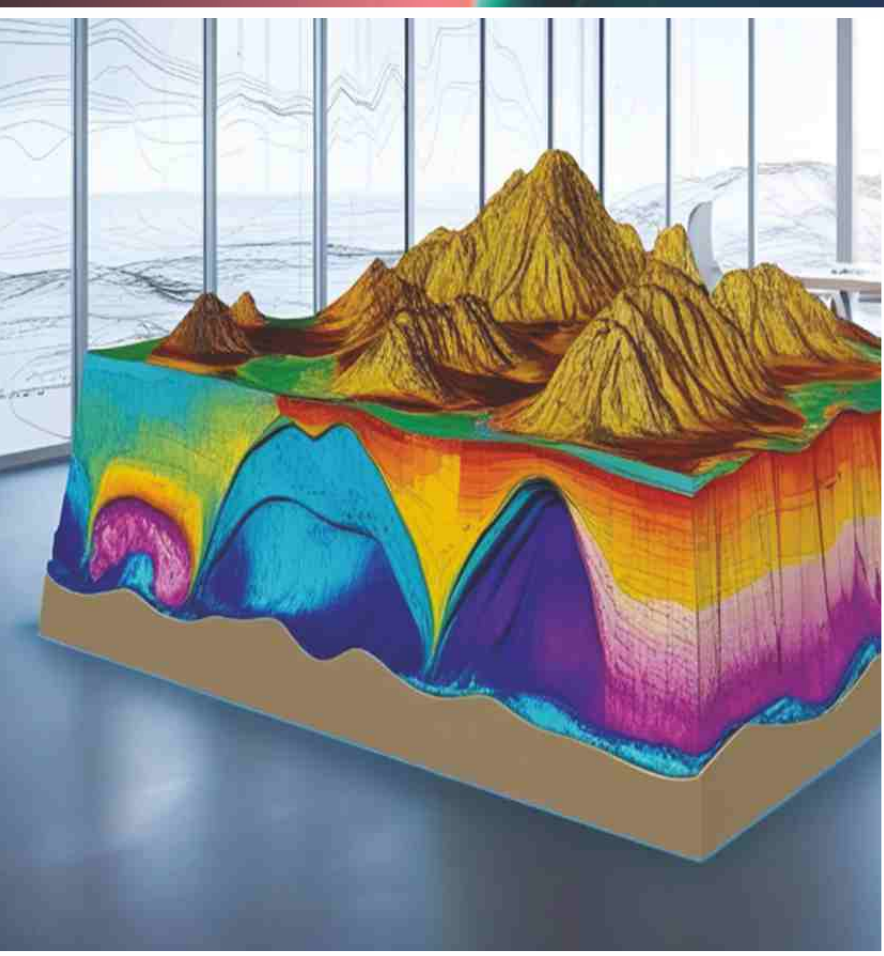
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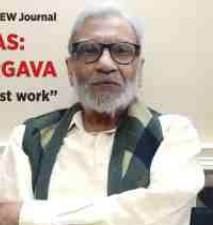
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Exploration Geologist, oil and gas industry, Dr. Sandip Kumar Roy has a wide exposure to Indian and International sedimentary basins. A recognized industry professional on geology of Andaman basin, India Dr. Roy has served ONGC Ltd for twenty one years followed by stints in BG Group (UK), Petra Energia (Brazil), Petronas (Malaysia) and the Directorate General of Hydrocarbons (DGH), Government of India. He holds a Ph.D degree in process Sedimentology of Andaman Flysch from Indian Institute of Technology, Bombay.

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IDENTIFYING A NEW OIL PRESENCE FROM GEOCHEMICAL ANALYSIS

Martin Fowler of Norway based Applied Petroleum Technology (APT) on his new study for Oil find

DEW EXCLUSIVE



Martin Fowler, Head, Applied Petroleum Technology's (APT) Canadian operation, has conducted a detailed evaluation of the Southern Grand Banks petroleum system and identified new oil presence offshore Canada. The study has for the first time demonstrated a correlation between Lower Jurassic source rocks in Europe and across the North Atlantic conjugate to Canada. It is obviously a big jump between identifying a new oil presence from geochemical analyses to making an actual discovery, but Applied Petroleum Technology believes that its analyses are of interest to energy companies that are investigating opportunities offshore Canada.

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Expert Talk: Devashish Chakravarty converse with Chief Editor, DEW Journal

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Why re-categorization of Indian Sedimentary Basins done as per PRMS?

Mr. Devashish Chakravarty, MTech in Petroleum Exploration from IIT-ISM, Dhanbad has over 35 years experience in various aspects of E&P Industry, specially in Strategy Development & Policy Formulation. He has wide experience of working with GoI on various upstream policy formulation. He recently superannuated as Chief General Manager from ONGC.



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Geological CO₂ storage in Basalt and unconventional formations

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This study assesses the potential for geological CO₂ storage in India's Deccan basalt and unconventional reservoirs like coal and shale. CSIR–National Geophysical Research Institute, with national and international partners, is conducting integrated petrophysical, geomechanical, geochemical, and seismic analyses, along with reactive transport modeling, to evaluate basalt formations. In the Raniganj Basin, coal and shale are being studied using core analysis, seismic data, and machine learning techniques. Advanced seismic modeling is also being developed for leakage detection. These efforts aim to support India's net-zero goals through secure and scalable CO₂ storage solutions.

Introduction

As India moves towards its ambitious goal of achieving net-zero carbon emissions by 2070, finding effective ways to reduce greenhouse gases is more important than ever. One promising solution is geological carbon sequestration, i.e., capturing CO₂ and storing it safely in underground geological formations like depleted oil and gas reservoirs, saline aquifers, mafic-ultramafic rocks, unmineable coal seams, and shale formations (Fig. 1)^[1].

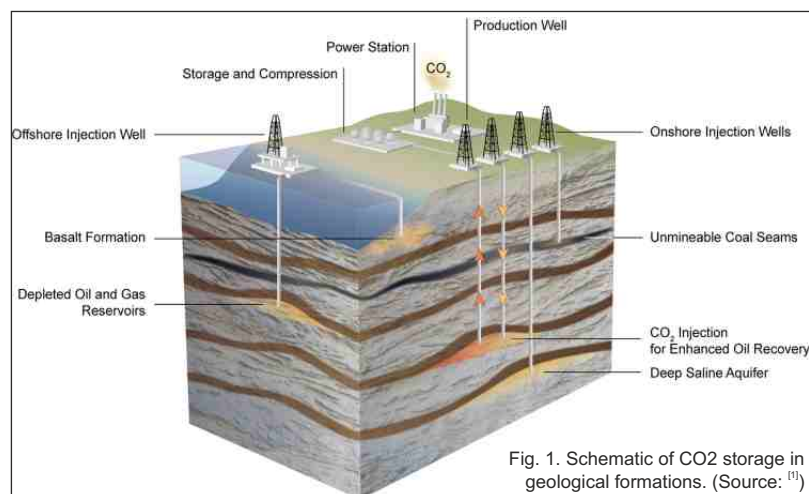


Fig. 1. Schematic of CO₂ storage in geological formations. (Source: ^[1])

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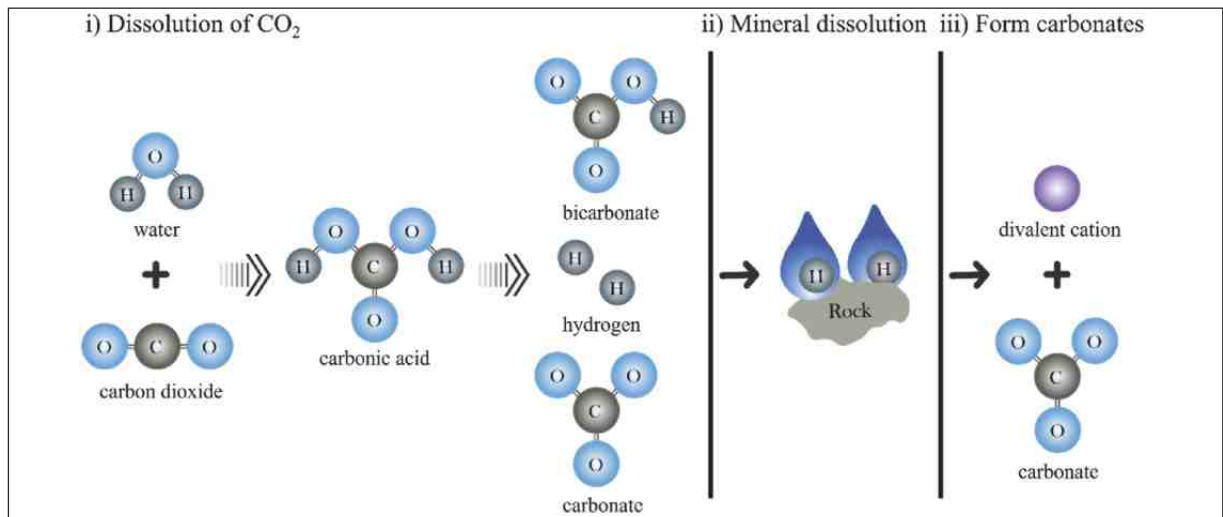


Fig. 2. Mineralization process for CO₂ storage in Basalt. (Source: [2])

The Shallow Seismic Group of CSIR–National Geophysical Research Institute (NGRI) is currently working on innovative methods to assess the feasibility of storing CO₂ in the basalts and coal-shale formations through an interdisciplinary approach that combines various observations, including geophysical, geochemical, remote sensing, and numerical modelling.

CO₂ storage in Basalt

Carbon storage in mafic/ultramafic rocks happens by virtue of chemical reaction between the aqueous or supercritical CO₂ and the divalent cations like calcium, magnesium, and iron present in feldspar, pyroxene, and olivine minerals.

This process effectively locks away carbon in the

reservoir for geological timescales. Unlike saline aquifers or depleted oil fields, where CO₂ remains in a supercritical or dissolved state with a risk of leakage, mafic/ultramafic rocks allow rapid and irreversible mineral trapping (Fig. 2). This approach has already been successfully demonstrated for existing projects like CarbFix in Iceland and Wallula in the United States for basalts. In India, many research groups are working on this topic, focusing on different aspects of the site suitability studies in the Deccan Traps. This flood basalt covers a vast area of over 500,000 sq km. CSIR-NGRI, in collaboration with the Indian Institute of Science Education and Research Bhopal (IISERB) and other scientific organisations, is leading an effort to assess the carbon storage potential of the Deccan mainland (<https://www.geomar.de/en/fb4-gdy/projects/translate-to-english-perbas>). A proof of concept has been shown in Fig. 3.

CSIR-NGRI, in collaboration with IISERB, is carrying out a comprehensive characterization of Deccan basalt flows for which the area is first narrowed down using key factors, viz. environmental, hydrological, geological, and economic factors, etc., to identify the most suitable sites for CO₂ storage. Thereafter, extensive fieldwork was conducted to collect rock samples for laboratory analysis to determine their capacity to safely and permanently

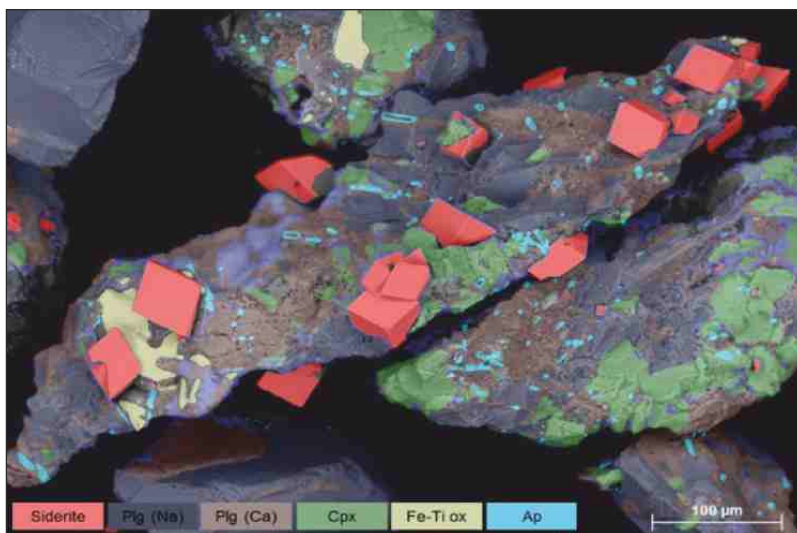


Fig. 3. False colour image of the basalt rock fragments showing siderite formation (red) after 81 days of reaction with CO₂. (Source: [2])

store CO₂. Core samples from the existing Killari borehole drilled by NGRI and the Atomic Minerals Directorate for Exploration and Research (AMD) were also included to examine the vertical variability within the basalt layers. These basalt samples will be used to determine the porosity, permeability, mineralogy, geomechanical strength, and geochemical reactivity. These properties are important to characterize the basaltic reservoir.

High porosity and permeability of a rock facilitate the storage and movement of reactive fluids through the rock matrix to a larger extent, which increases the surface area for the reaction of CO₂ with minerals. This, in turn, also reduces the pressure in the reservoir and lowers the risk of induced seismicity. In this study, porosity was measured using two methods- helium porosimeter and high-resolution X-ray computed tomography (CT). Permeability was estimated through a pore-connectivity map (using high-resolution CT images). It was observed that the results obtained from all these methods vary. In case of rocks like basalt, higher resolution techniques are required.

Rock mechanical properties give us a cue about the formation's ability to withstand high pressure. For example, high injection pressures can create fractures, and the formation may lose its structural integrity, making it prone to leakage. Similarly, it is important to assess the rock mechanical properties post-CO₂ injection. For example, dissolution of cation-bearing minerals and subsequent carbonate mineralization can significantly alter rock strength. Thus, geomechanical characterization is a critical component of the feasibility assessment. It also plays a key role in ensuring the long-term integrity of the caprock. Geomechanical investigations, such as in-situ triaxial tests, ultrasonic velocity tests, uniaxial compression, direct and indirect tensile tests, and scratch tests, were conducted. These tests help in estimating the elastic moduli of the samples and provide a comprehensive understanding of the mechanical behaviour of the formation. These studies are important for any injection project.

Mineralogical studies are the backbone of a permanent CO₂ storage project. Thus, in this study, in addition to standard petrographic studies, Scanning Electron Microscopy (SEM) was carried out to investigate micro-textures and alteration features. X-ray Fluorescence (XRF) and X-ray Diffraction (XRD) techniques were used to quantify major oxides and identify mineral phases favourable for carbonation.

Further, to simulate and monitor basalt–CO₂ interactions, batch reaction experiments were conducted under in-situ pressure and temperature conditions. These experimental results were further supported by simulations using reactive transport models such as PHREEQC, which were calibrated using the physical reaction data. Currently, quantification of the petrophysical and geochemical changes in the reacted core samples is going on to assess the extent of mineral transformation to understand the CO₂ storage efficiency.

In any geological storage case, sub-surface modelling plays a vital role; however, in basalt, the seismic imaging is very challenging, hence integrated geophysical studies are highly recommended. An acceptable geological model can be built using multi-parametric surface geophysical, borehole, and remote sensing data. Thus, potential injection zones can be identified by integrating all these information ensuring that the proposed site is both geologically suitable and environmentally secure.

CO₂ storage in Coal and Shale: Raniganj Basin

The Raniganj Basin, located in the easternmost part of the Damodar Valley Basin, hosts extensive reserves of high-volatile bituminous coal in the Raniganj and Barakar Formations interbedded with the Barren Measures Shale Formation. This geological setting makes the region a prime target for unconventional resource exploration. CSIR-NGRI is actively exploring the feasibility of CO₂ storage in unconventional reservoirs such as the unmineable coal seams and shale layers. Coal beds store CO₂ primarily through adsorption within their microporous matrix, making them highly suitable for CO₂ storage. Owing to its stronger affinity for coal surfaces compared to methane, CO₂ effectively displaces CH₄, and gets trapped within the coal micropores while simultaneously enhancing methane production, aiding Enhanced Coal Bed Methane (CBM) recovery as CBM is being actively produced from the Raniganj Basin ^[4]. In case of shale formations, it offers multiple mechanisms for CO₂ storage, including adsorption onto organic matter (such as kerogen) and clay minerals, and free gas accumulation in nano- to micropores. Their high surface area, especially in organic-rich zones with elevated Total Organic Carbon (TOC), enables substantial CO₂ adsorption, often exceeding that of methane due to CO₂'s higher molecular affinity (Fig. 4) ^[5,6].

In this direction, CSIR-NGRI, in collaboration with

CSIR-Central Institute of Mining and Fuel Research (CIMFR), Central Mine Planning & Design Institute (CMPDI), has evaluated the shale formation of the Damodar Basin of India. CSIR- NGRI conducted 3D seismic studies in the Rangamati area, located in the north-eastern part of the Raniganj Basin. The seismic data was acquired using a HEMI-50 vibrator with a 50000-pound peak force, with operating frequencies ranging from 10 to 150 Hz. In addition to seismic studies, geophysical well-logs and core analyses were carried out by CMPDI and CIMFR. The findings from this multidisciplinary study highlight the potential of Barren Measures Shale for both shale gas exploration and CO₂ storage. As the coal and shale formations in the Rangamati block are relatively shallow, further studies were carried out in the south-eastern block.

In the SE block, 2D seismic reflection and refraction, 2D three-component (3C) seismic reflection, and gravity-magnetic studies were carried out in collaboration with Essar Oil and Gas Exploration and Production Limited (EOGEL). Twelve multicomponent seismic reflection profiles of approx. 2.4 sq. km length, refraction profiles of 500 m length were shot to map coal seams, shales, and structural features. We observed that due to high seismic attenuation, the energy released by a single vibrator was not sufficient to illuminate the deeper targets. The near-surface noise was also high; however, multiple coal seams, structural discontinuities, and Barren Measures

shales were delineated in the seismic reflection studies, validated with the well-log data. The shallow refraction results revealed a three-layer structure in the shallow subsurface of the study area, which will be used to study the near-surface overburden. To examine anisotropic effects and direction-based variations in seismic responses, a targeted 2D-3C experiment was conducted. 2D-3C profiles were acquired in a star pattern in an increased seismic bandwidth of 10 to 200 Hz to test the anisotropic character of the rock that poses challenges in the quantification of resources based on seismic data. In this experiment, the receivers were laid in a circular pattern for full azimuthal coverage in an area of about 650m × 650m due to limited logistics. The survey layout resembles a pseudo-3D layout, and the source moved along the four cross lines in N-S, E-W, NE-SW, and NW-SE directions. However, it was observed that longer acquisition profiles were required for anisotropy characterization. Given the limited accessibility of the area, ideal seismic acquisition was constrained. Therefore, to further investigate the deeper sediments and features, such as intrusions, gravity, and magnetic studies were carried out. Gravity data were acquired along 21 traverses, and the results revealed a low gravity anomaly in the central-western part of the south-eastern Raniganj Basin, indicative of a deeper basin segment.

Machine Learning Based Shale Characterization

The shale-dominated Barren Measures Formation of the Raniganj Basin, though rich in organic matter, currently lacks a high-resolution, predictive framework. Conventional methods often fail to capture the spatial and compositional variability of shale properties, highlighting the need for a more robust, data-driven approach. To address this, we developed a multilayer perceptron (MLP) neural network capable of modeling complex, non-linear relationships between geophysical well logs and laboratory-measured shale properties^[7]. Trained on core-derived data, the model predicts critical parameters such as total organic carbon (TOC), pyrolyzable shale hydrocarbons (S₂), hydrogen index

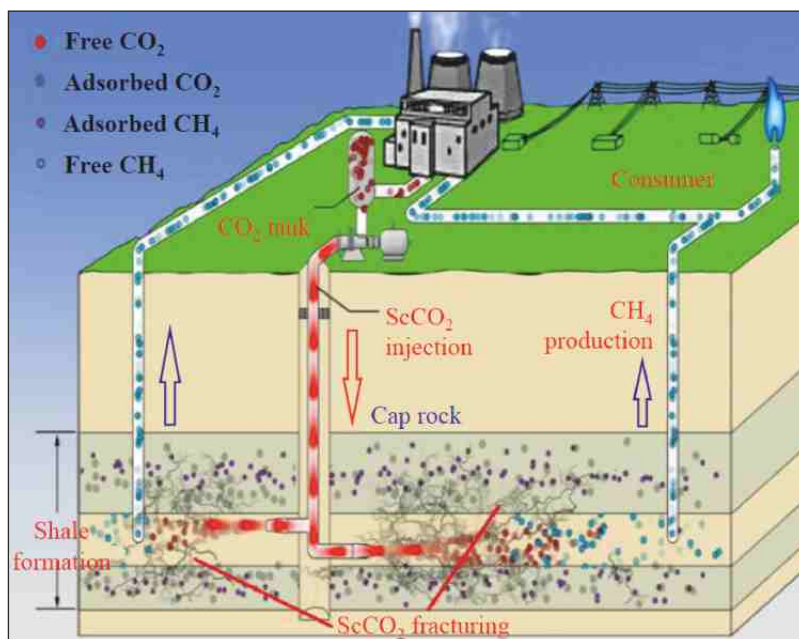


Fig. 4. Supercritical CO₂ based fracturing technology for CO₂ sequestration and enhancing shale gas recovery (Source: ^[6])

(HI), and kerogen type (Fig. 5).

These outputs are integrated with 3D seismic, mineralogical, and geomechanical datasets to enable a comprehensive assessment of shale reservoir quality. Building on this, our future work aims to integrate seismic attributes with petrophysical and geochemical data using advanced machine learning models, such as convolutional neural networks (CNNs), to further improve the spatial resolution of shale characterization and identify optimal targets for CO₂ storage.

Developments in Reservoir Characterization Workflows

For reservoir characterization, high-resolution geomodels are a prerequisite. Hence, to improve the resolution, we employed Joint Refraction and Reflection (JRR) seismic tomography to construct a detailed velocity model. This approach allows for the detection of thin coals and intercalated shale layers, which are often missed by conventional processing methods. We also developed techniques to enhance the spectral bandwidth of

the data by developing BPI-based inversion^[8] and have developed a user-friendly software package, 'SeisImage', designed for seismic inversion. This GUI-based tool integrates various geophysical datasets and

applies fractional Gaussian-based stochastic inversion methods, leveraging fractal properties to improve resolution. Its robustness against noise enables the detection of very thin layers (Fig. 6).

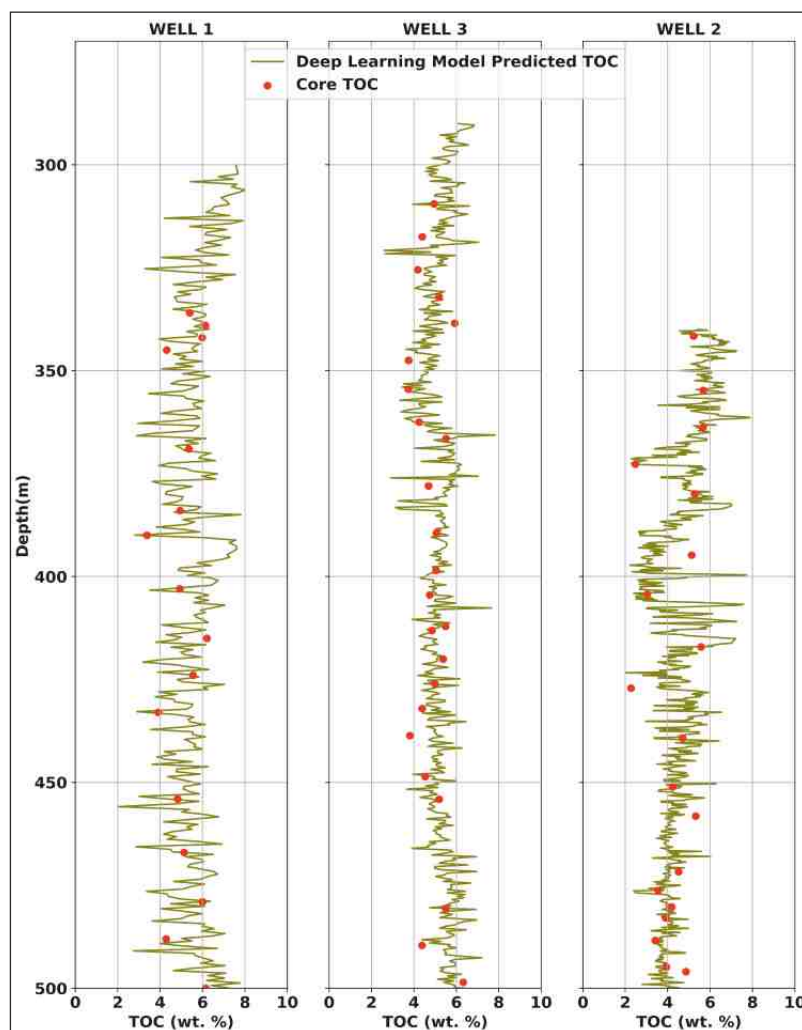


Fig. 5. Depth-wise comparison of deep learning-predicted TOC (olive lines) with core-derived TOC data (red dots) across three wells. (Source: ^[7])

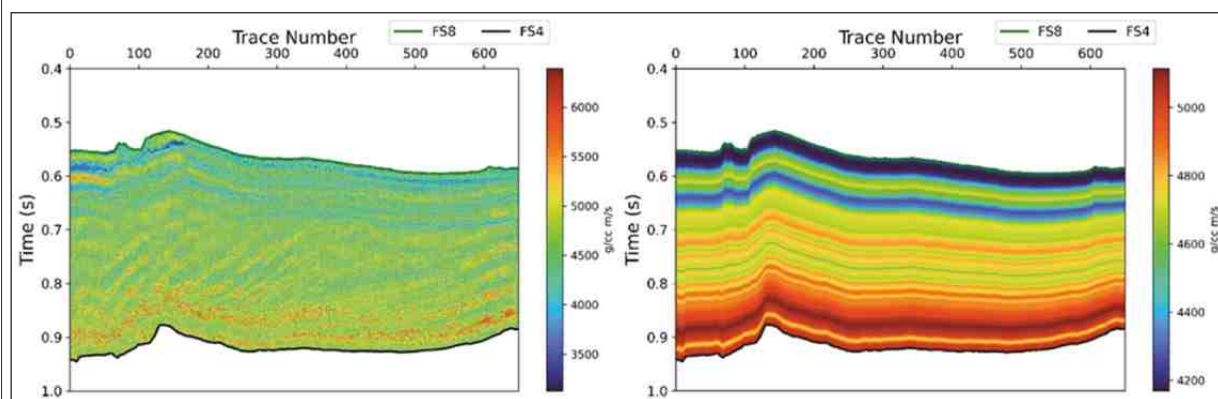


Fig. 6. The seismic data from the F3 block, Netherlands, is used for demonstration of the inversion process. (left) Low-frequency impedance model generated for the F3 data, (right) the final impedance model was obtained after carrying out the inversion algorithm. (Source: ^[9])

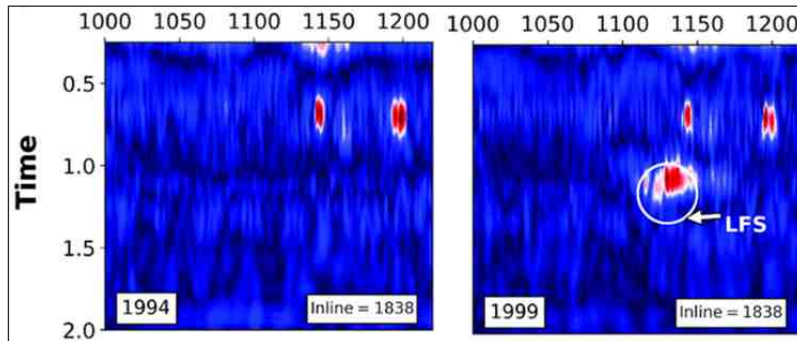


Fig. 7. Common frequency section along an inline of Sleipner crossing the CO₂ injection point for 1994 (top), 1999 (bottom). The circle represents a low-frequency shadow at 12 Hz. (Source: ^[10])

Ongoing Research on Seismic Monitoring

Building on our previous work, including the simulation of CO₂ migration in brine-saturated Ardol sand (Cambay Basin) using axisymmetric gravity currents and poroviscoelastic models, as well as the development of a seismic monitoring workflow based on low-frequency shadows applied to monitor CO₂ movement in the Sleipner field (Fig. 7) ^[10], we further aim to extend these methodologies to future pilot injection sites. Our modeling studies have demonstrated that the viscosity and diffusivity of injected CO₂ play an important role in its movement. As part of our future work, we plan to refine detection thresholds for small-scale leakages, enhance the sensitivity of seismic monitoring tools, and integrate these approaches into a robust, site-specific monitoring framework. Additionally, we will focus on improving subsurface models and evaluating the energy efficiency and cost-effectiveness of geological CO₂ storage (GCS) in the Indian context to support safe, scalable, and economically viable deployment.

Concluding Remarks

This study discusses the potential of geological CO₂ storage in India's Deccan basalt and unconventional formations such as coal and shale. Basalts offer long-term mineral trapping, while coal and shale formations primarily provide adsorption-based storage with possible cleaner methane recovery. Multidisciplinary geoscientific analyses, including seismic imaging, geomechanical testing, geochemical experiments, and machine learning models, have enhanced reservoir characterization and site suitability assessment. Further advanced seismic workflows and monitoring tools are being developed for small leakage detection. These integrated efforts support the establishment of safe, scalable CO₂ storage strategies aligned with India's net-zero targets and lay the

groundwork for future pilot injection and monitoring projects.

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image acquisition, geomechanical tests, and batch reactions of the basalt samples. The petrophysical and geomechanical properties of the coal and shale formations, including permeability and strength, have been assessed in collaboration with KAUST. The authors acknowledge all the collaborators, including IISERB, CSIR-CIMFR, CMPDI, and EOGEP.

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